rdfia 3-a

1. We need the boundary conditions $\alpha_0 = 1$, $\beta_0 = 0$ and $\alpha_1 = 0$, $\beta_1 = 1$

With $\alpha_t = \cos(at)$, $\beta_t = \sin(bt)$:

- $\alpha_0 = \cos(0) = 1$ ok ; $\beta_0 = \sin(0) = 0$ ok
- $\alpha_1 = \cos(a) = 0 \Rightarrow a = \frac{\pi}{2}$
- $\beta_1 = \sin(b) = 1 \Rightarrow b = \frac{\pi}{2}$

So $a = b = \frac{\pi}{2}$

2. $X_t = \alpha_t X_0 + \beta_t X_1 = \cos(at) X_0 + \sin(bt) X_1$.

Hence, $\frac{dX_t}{dt} = \alpha_t' X_0 + \beta_t' X_1 = -a\sin(at) X_0 + b\cos(bt) X_1$.

With $a = b = \frac{\pi}{2}$: $\frac{dX_t}{dt} = -\frac{\pi}{2}\sin\left(\frac{\pi t}{2}\right) X_0 + \frac{\pi}{2}\cos\left(\frac{\pi t}{2}\right) X_1$.

3. $X_{\frac{1}{2}} = \frac{1}{2} X_0 + \frac{1}{2} X_1$

$\mathrm{Var}(X_{\frac{1}{2}}) = \frac{1}{4}\mathrm{Var}(X_0) + \frac{1}{4}\mathrm{Var}(X_1) = \frac{1}{4} \times 1 + \frac{1}{4} \times 1 = \frac{1}{2}$.

$\mathrm{Var}(X_0) = \mathrm{Var}(X_1) = 1 \neq \mathrm{Var}(X_{\frac{1}{2}})$

So the schedule is not variance preserving.

4. We suppose that $\mathrm{Var}(X_0) = \mathrm{Var}(X_1) = 1$ and $X_0 \perp\!\!\!\perp X_1$.

$\mathrm{Var}(X_t) = \cos^2\left(\frac{\pi t}{2}\right) \mathrm{Var}(X_0) + \sin^2\left(\frac{\pi t}{2}\right) \mathrm{Var}(X_1)$

$= \cos^2\left(\frac{\pi t}{2}\right) + \sin^2\left(\frac{\pi t}{2}\right) = 1$.

So the schedule is variance preserving.

5. A variance-preserving schedule keeps the intermediate distributions at roughly the same "scale" as the endpoints, so the trajectory from μ_0 to μ_1 does not collapse (too small variance) or blow up (too large variance).

Geometrically, samples stay on a tube of comparable radius instead of

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shrinking toward the mean and then re-expanding. This makes the vector field easier to learn and the ODE integration more stable : the network sees inputs with consistent magnitudes across t, avoids extreme signal-to-noise regimes, and typically yields smoother paths and better numerical conditioning during generation.

6 Yes, the experimental observations clearly confirm the theoretical predictions. With the linear schedule, the trajectories visibly contract toward the mean at intermediate times, which indicates a reduction of variance during the transport. This matches the theoretical result that linear interpolation is not variance preserving.

In contrast, with the cosine schedule, the trajectories remain well spread throughout the entire evolution. The variance of the generated samples stays approximately constant at each time step, which is consistent with the theoretical property that the cosine schedule is variance preserving.

Computing the empirical variance along time further confirms this behaviour: it decreases for the linear schedule but remains stable for the cosine schedule.

7. The main advantage of Diffusion models and Flow Matching is training stability and coverage of the data distribution.

They rely on a well-defined regression objective instead of adversarial training, which avoids common GAN issues such as mode collapse and unstable optimization. As a result, they typically produce more diverse and reliable samples.

8. The main computational disadvantage is that generation is iterative Diffusion models and Flow Matching require solving a sequence of steps

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to generate one sample, whereas GANs generate samples in a single forward pass. This makes sampling slower and more computationally expensive.

9. Although Flow Matching generates images in multiple steps, each step models a simpler transformation. The generation process is decomposed into many small, well-conditioned updates, which allows the model to capture complex data distributions more accurately.

In contrast, GANs must learn the entire transformation in a single step, which is often harder and often leads to instability or missing modes. This multi-step refinement explains why Flow Matching typically yields higher-quality and more diverse samples given a similar number of parameters.

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